A Discrete Causal Lattice Foundation for Physics: The Octahedral $\mathcal{A} = 1$ Framework

Jack Menendez

Eugene, OR

ORCID: 0009-0003-1166-307X

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Abstract

[STUB] The abstract will summarize the A=1 framework applied to a three-dimensional octahedral causal lattice $\mathcal{T}_\diamond^3$, in which every particle carries an independent tick counter and physics emerges from the combinatorial scheduling of those counters. Gravity arises as a clock-density gradient rather than spacetime curvature. A falsifiable prediction concerning discrete path-count deviations from the continuous $1/r^2$ law is derived and proposed for experimental verification.

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1 Introduction

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2 The Octahedral Substrate ($\mathcal{T}_\diamond^3$)

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6 Emergent Kinematics: Momentum and Force in the $\mathcal{T}_{3d}$ Manifold

7 Emergent Kinematics on the $\mathcal{T}_\diamond^3$ Lattice

Momentum, inertia, and force are not postulated in the $\mathcal{A} = 1$ framework. They emerge from the geometry of phase interference across the bipartite sublattice structure. This section derives the kinematic relations from the Dirac tick rule and connects each result to the corresponding numerical verification in the experiment suite.

7.1 The Bipartite Velocity

A massive Dirac fermion [1] on $\mathcal{T}_\diamond^3$ carries two spinor components: ψ_R on the RGB sublattice (even ticks) and ψ_L on the CMY sublattice (odd ticks). The $\mathcal{A} = 1$ constraint requires

$$\sum_{\mathbf{x}} \left(|\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2 \right) = 1. \tag{1}$$

Over one complete two-tick cycle the kinetic fraction $p_{\text{hop}} = \cos^2(\omega/2)$ is distributed across the six basis vectors, split equally between the three RGB directions ($\mathbf{V}_{1\text{–}3}$, even tick) and the three CMY directions ($-\mathbf{V}_{1\text{–}3}$, odd tick):

$$1 = p_{\text{stay}} + P_{\text{fwd}} + P_{\text{bwd}}, \qquad p_{\text{stay}} = \sin^2\tfrac{\omega}{2}, \quad P_{\text{fwd}} + P_{\text{bwd}} = \cos^2\tfrac{\omega}{2}. \tag{2}$$

Here $P_{\text{fwd}} = \sum_{i=1}^{3} p(\mathbf{V}_i)$ is the total probability flowing along the RGB sublattice and $P_{\text{bwd}} = \sum_{i=1}^{3} p(-\mathbf{V}_i)$ along CMY. The lattice speed limit $c = 1$ node/tick gives the net velocity

$$v = c\,(P_{\text{fwd}} - P_{\text{bwd}}). \tag{3}$$

For a packet at rest the two sublattices are populated symmetrically, $P_{\text{fwd}} = P_{\text{bwd}} = \frac{1}{2}\cos^2(\omega/2)$, so $v = 0$ and the amplitude oscillates between ψ_R and ψ_L without net displacement — the lattice realisation of Zitterbewegung [2].

7.2 Phase Gradient as Momentum

A uniform spatial phase gradient with advance θ per lattice edge breaks the RGB/CMY symmetry. In the implementation this gradient appears as the per-direction phase advance

$$\delta_p(\mathbf{V}_i, \mathbf{x}) = \arg\psi(\mathbf{x} + \mathbf{V}_i) - \arg\psi(\mathbf{x}), \tag{4}$$

which is the quantity `delta_p` computed in `CausalSession._kinetic_hop`. Only positive advances contribute to the hop weight, so amplitude flows preferentially in the direction of increasing phase — the lattice realisation of the de Broglie relation [3].

To derive $v(\theta)$ analytically, consider a packet with mass parameter ω (the instruction frequency) subject to a gradient θ directed along the net RGB axis. On the even (RGB) tick ψ_R hops forward: the phase gradient $+\theta$ reduces the effective mismatch from ω to $\omega - \theta$, increasing the kinetic weight. On the odd (CMY) tick ψ_L hops backward: the reversed

gradient adds $+\theta$ to the mismatch, decreasing the kinetic weight:

$$P_{\text{fwd}} = \tfrac{1}{2}\cos^2\tfrac{\omega-\theta}{2}, \qquad (\psi_R \text{ hop, even tick}) \tag{5}$$

$$P_{\text{bwd}} = \tfrac{1}{2}\cos^2\tfrac{\omega+\theta}{2}. \qquad (\psi_L \text{ hop, odd tick}) \tag{6}$$

Substituting into Eq. (3) and applying $\cos^2\alpha - \cos^2\beta = -\sin(\alpha+\beta)\sin(\alpha-\beta)$ yields

$$v = \tfrac{c}{2}\,\sin(\omega)\,\sin(\theta). \tag{7}$$

Table 1 illustrates how p_{stay}, m_{inertial}, and the resulting velocity vary across representative values of ω: In the macroscopic limit $\theta \ll 1$ (gradient small compared to 2π), $\sin\theta \approx \theta$.

ω	$p_{\text{stay}} = \sin^2(\omega/2)$	$m_{\text{inertial}} = \sin(\omega)/2$	v/c at $\theta = 0.1$
0	0	0	1 (photon limit)
$\pi/4$	0.146	0.354	0.035
$\pi/2$	0.500	0.500	0.050
π	1.000	0	0 (frozen, max mass)

Table 1: Selected values of the two mass-related quantities and the resulting lattice velocity at a small phase gradient $\theta = 0.1$. The photon row ($\omega = 0$) is handled by the massless limit (Section **??**); the other rows follow Eq. (7).

Identifying the lattice momentum $p \equiv m\,v/c$ and the effective mass $m \equiv \sin(\omega)/2$[1] recovers the classical kinematic relation

$$p = m\,\theta \cdot c, \tag{8}$$

where θ is the phase advance per lattice edge — the discrete analogue of the wavenumber k. Momentum is thus strictly the cross-term between the particle's internal mass-phase ω and the spatial phase gradient of the bipartite vacuum.

Massless limit. Equation (7) gives $v = 0$ when $\omega = 0$, which is not the photon behaviour. The massless case is qualitatively distinct: with $\omega = 0$ the residence probability vanishes, $p_{\text{stay}} = 0$, and the amplitude hops entirely via the directional weighting of Eq. (4). The hop is strictly one-way along the gradient ($\delta_p > 0$ only), so $P_{\text{bwd}} = 0$ and $v = c$ regardless of θ. The two-sublattice derivation above applies to massive particles ($\omega > 0$); the photon is the $\omega \to 0$ limit treated separately in Section **??**.

[1] The inertial mass $m_{\text{inertial}} = \sin(\omega)/2$ is *not* the same as the residence probability $p_{\text{stay}} = \sin^2(\omega/2)$. They coincide only at $\omega = \pi/2$, the unique frequency at which the particle spends exactly half its time at rest. At $\omega = \pi$ (maximum residence) the particle is frozen, $m_{\text{inertial}} = 0$, while $p_{\text{stay}} = 1$.

Numerical verification. Experiment `exp_01_inertia.py` initialises a Gaussian wavepacket ($\sigma = 3$, $\omega = 0.2$) with momentum $\mathbf{k} = (k, k, k)$, $k = 0.3/\sqrt{3}$, corresponding to a phase advance $\theta = \mathbf{k}\cdot\mathbf{V}_1 = 0.3\sqrt{3} \approx 0.520$ rad per hop along the $\mathbf{V}_1 = (1, 1, 1)$ direction. Equation (7) predicts

$$v_x = \tfrac{1}{2}\sin(0.2)\sin(0.3\sqrt{3}) \approx 0.0493 \text{ nodes/tick}.$$

The measured x-component fit over 12 ticks gives $v_x = 0.0464$ nodes/tick — a 6% discrepancy attributable to finite packet width and the short run length. The heavy particle ($\omega = 0.5$, same $\mathbf{k}$) drifts measurably slower, confirming that $\sin(\omega)$ governs inertial resistance: a larger phase cost per hop reduces the net kinetic probability imbalance.

7.3 Acceleration and Force

Acceleration requires a time-varying gradient. Taking the discrete time derivative of Eq. (7) with ω held constant:

$$a = \frac{\Delta v}{\Delta t} = \frac{c}{2}\,\sin(\omega)\,\cos(\theta)\,\frac{\Delta\theta}{\Delta t}. \tag{9}$$

For small gradients ($\cos\theta \approx 1$) this reduces to the lattice analogue of Newton's second law [4]:

$$a = \frac{c}{2m}\,F_\theta, \qquad F_\theta \equiv \frac{\Delta\theta}{\Delta t}. \tag{10}$$

The quantity F_θ — the rate at which the spatial phase gradient changes per tick — is the lattice definition of force. A gravitational or electromagnetic field acts by twisting the phase of the vacuum over time, unbalancing P_{fwd} and P_{bwd} and continuously converting p_{stay} into directed kinetic probability. The mechanism by which the Coulomb potential produces F_θ in the hydrogen orbital is analysed in Section **??**.

8 Gravity as Clock Density

[STUB]

9 Clock Fluid Dynamics and the Continuum Limit

[STUB – see notes/clock_fluid_dynamics.md]

10 The Vacuum Twist: Gravity, Electromagnetism, and the Unified Phase Field

[STUB – see notes/vacuum_twist_field_equations.md]

11 Interference and the Huygens Lantern

[STUB]

12 The Observer as Clock

13 The Observer as a Clock

In standard quantum mechanics the measurement problem is left philosophically unresolved. Copenhagen says: before measurement, a particle is in a superposition; measurement collapses it to a definite state instantaneously, by an unspecified mechanism, at an unexplained boundary between quantum and classical [5]. The mechanism of collapse is not derived from the formalism — it is postulated as an axiom of interpretation.

The $\mathcal{A} = 1$ framework replaces this postulate with a precise mechanical statement: *an observation is the event in which a new clock session is registered in the* `TickScheduler`. There is no collapse, no special role for consciousness, and no quantum/classical boundary. The same tick rule that governs particle propagation governs the observer. The asymmetric effect we call "measurement" follows from the asymmetry in coherence length between the two cones involved.

13.1 Observation as Clock Registration

Every physical system in the framework is a `CausalSession`: a normalised spinor (ψ_R, ψ_L) evolving under the Dirac tick rule on the $\mathcal{T}_\diamond^3$ lattice, carrying an instruction frequency ω and a tick counter maintained by the `TickScheduler`. When two sessions are registered in the same scheduler, the scheduler applies pairwise phase interactions at shared nodes after each tick — the *clock entanglement* that constitutes observation.

The state space of the scheduler is the set of all orderings of its registered clocks. For n clocks this has size $n!$:

$$|\Omega_n| = n!, \tag{11}$$

verified numerically in `exp_05`: $1! = 1$, $2! = 2$, $3! = 6$, $\ldots$, $6! = 720$, each step monotonically larger than the last. The key consequence of Eq. (11) is that the state space grows with every new registration and never shrinks: there is no `remove_session` method on `TickScheduler` by design. Observation is architecturally irreversible.

The probability constraint $\mathcal{A} = 1$ is preserved regardless of how the clocks are ordered. All three shuffle schemes tested in `exp_05` — sequential, random, and priority — maintain $|\sum_i \rho_i - n| < 10^{-15}$. The physics does not depend on which observer goes first; only the correlations between their futures depend on the ordering.

13.2 Asymmetric Phase Exchange and Apparent Collapse

When an observer session and a particle session co-exist in the same scheduler, their cones overlap at shared nodes. The `TickScheduler` applies a phase rotation to both sessions at each overlap, mixing the phase structures of the two cones. After the interaction each cone contains a modified amplitude and phase distribution — not randomly, but specifically, determined by the phase information that leaked from one cone into the other.

This is not wavefunction collapse. Nothing is destroyed. Each cone still exists, still satisfies $\mathcal{A} = 1$, still encodes a full distribution over possible futures. What changed is *which futures are constructively amplified.*

The asymmetry that produces the apparent collapse is a consequence of the relative coherence lengths of the two cones. A freshly prepared particle carries a narrow, precisely structured interference pattern: most of its probability is concentrated on a small number of paths, and the phase relationships between those paths are what produces the interference fringes. An observer — a macroscopic detector or any sufficiently large coupled system — carries a broad, incoherent cone: many degrees of freedom, already deeply mixed, with phases effectively random relative to each other.

When they overlap, phase information flows in both directions. But the effect is asymmetric. The observer's cone is so large and already so incoherent that the small phase perturbation from the particle makes essentially no difference to its future: the observer's possible futures are barely changed. The particle's cone is small and precisely coherent. The phase perturbation from the observer's broad incoherent field is large relative to the particle's existing phase structure. The particle's interference pattern is scrambled significantly — its possible futures are substantially redistributed.

`exp_05` Test 3 demonstrates this directly. A particle session ($\sigma = 3$, $\omega = 0.2$) propagates freely for 10 ticks and accumulates a density ρ_{free}. The same particle evolving alongside an observer session ($\sigma = 1.5$, $\omega = 0.5$, broader frequency, narrower spatial extent) accumulates

a density ρ_{obs}. Both satisfy $A = 1$ to machine precision ($|\sum \rho - 1| < 10^{-8}$), but

$$\langle |\rho_{\text{free}} - \rho_{\text{obs}}| \rangle = 1.60 \times 10^{-4},$$

confirming that the observer altered the particle's future without violating probability conservation.

The mechanism of apparent collapse. Before the observer interaction, the particle's cone encodes a superposition of futures — say, "goes left" and "goes right" with equal amplitude and a specific phase relationship between them. That phase relationship is what produces the interference pattern. After the observer interaction, the phase relationship between "goes left" and "goes right" has been scrambled. The amplitudes are the same, but the phases are now random relative to each other. Destructive interference no longer suppresses either path — the interference pattern disappears.

From the outside this looks like collapse: the particle "chose" left or right. What actually happened is that the phase information encoding the interference between left and right was transferred into the observer's cone, where it is diluted among the observer's vast number of degrees of freedom and effectively inaccessible. The future that was being suppressed by destructive interference is now equally probable, not because it was created, but because the information suppressing it was removed.

The interaction between an observer session and a particle session modifies the interference structure of the particle's causal cone — the amplitude distribution encoding the particle's possible futures. Because the observer's cone carries many more degrees of freedom than the particle's cone, the phase exchange is asymmetric: the observer's future is negligibly modified while the particle's future is substantially restructured. What appears as wavefunction collapse in the Copenhagen interpretation is, in the $\mathcal{A} = 1$ framework, precisely this asymmetric phase exchange between cones of vastly different coherence length. No information is destroyed; it is redistributed from the particle's coherent cone into the observer's incoherent one, where it becomes inaccessible rather than absent.

13.3 Irreversibility and Entropy

The irreversibility of measurement is, in the $\mathcal{A} = 1$ framework, an architectural fact rather than a thermodynamic approximation. `TickScheduler` has no `remove_session` method. Once a clock is registered the state space grows from $(n-1)!$ to $n!$, and this cannot be undone.

This gives a precise lattice definition of entropy increase: the entropy of the scheduler's

state space is $\log|\Omega_n| = \log(n!)$. Each observation adds one clock and increases the entropy by $\log n$ — the number of ways the new clock can be interleaved with the existing $n-1$ clocks. The second law of thermodynamics, in this picture, is the statement that observation is irreversible clock registration: you cannot un-observe a particle.

No special role for consciousness. Any physical system that registers as a `CausalSession` in the scheduler is an observer. A photon detector, a dust grain, a passing cosmic ray — any interaction that entangles two tick counters constitutes a measurement. Consciousness is not a prerequisite; it is merely one instance of a large-cone incoherent system interacting with a small-cone coherent one. The difference between a measurement and a controlled coherent interaction is only one of the structure of the observer's cone: incoherent cones scramble interference patterns; carefully prepared coherent cones can impose them.

13.4 Entanglement as Correlated Cone Structure

When two cones overlap and exchange phase in the `TickScheduler`, they become correlated: the phase structure of each cone now contains information about the other. This is entanglement, described exactly and locally.

Subsequent modification of one cone — by a further observation — propagates into the correlated phase structure. The second cone still contains the phase information from the first exchange, so its future is correlated with the outcome of the second measurement on the first particle. This correlation is not non-local in the sense of faster-than-light signalling. The correlation was established when the cones overlapped, in a local interaction bounded by c. The subsequent measurement travels at c within each cone. No information moves faster than the cone boundary, but the pre-established correlation means two widely-separated cones can have correlated futures even without recent interaction — the $\mathcal{A} = 1$ account of Bell correlations [6].

14 Path Counting and Falsifiable Predictions

[STUB]

15 Lattice Harmonics and Quantization

[STUB – see notes/lattice_harmonics.md and exp_09_harmonics.py]

16 The Hydrogen Spectrum from Lattice Geometry

[RESULT: r_2/r_1 = 4.42, E_2/E_1 = 0.229. Bohr spectrum confirmed at 8-10%. Spontaneous n=2 to n=1 decay observed at tick 340. Both results are emergent.]

17 System Audit and Traceability Matrix

Table 2: System Audit: Mapping $\mathcal{T}^3_\diamond$ Protocols to Experiments

Physical Phenomenon	Lattice Logic	Manifold Limit	Experiment	Status
Speed of Light Limit	Topological Adjacency ($v_{max} = 1$)	Planck Ratio ($c = l_P/t_P$)	exp_00_causal_cone.py	PASS
Inertial Persistence	Phase Gradient ($\nabla\phi$)	Newton's First Law	exp_01_inertia.py	PASS
Gravitational Pull	Clock Density Gradient	Schwarzschild Metric	exp_02_gravity_clock_density.py	PASS
Quantum Interference	Discrete Path Summation	Schrödinger Equation	exp_03_interference.py	PASS
Wave Function Collapse	Observer Clock Addition	Density Matrix Decay	exp_04_decoherence.py	PASS
Measurement Problem	Tick Sequence Entanglement	Decoherence Theory	exp_05_observer_clock.py	PASS
$1/r^2$ Correction	Discrete Path Count Deviation	Gaussian CLT Limit	exp_06_path_counting.py	PASS
Time Dilation Quantum	Minimum Clock Increment	Schwarzschild Time Dilation	exp_06_path_counting.py	PASS
Clock Density Conservation	Continuity Equation from tick()	Einstein Field Equations	exp_07_clock_conservation.py	PASS
Gravity vs EM Twist & div(ϕ) vs curl(ϕ) & Maxwell + Einstein & exp_08_vacuum_twist.py & STUB				
Zitterbewegung Spectrum	Frequency scan, stability peaks	Mass quantization	exp_09_harmonics.py	PASS
Orbital Resonance	Stable orbits in clock-density well	Bohr spectrum	exp_09_harmonics.py	PART
Photon Dispersion	Group velocity vs wavenumber k	Brillouin zone boundary	exp_09_harmonics.py	PASS
Temporal Locking	Frequency drift toward rationals	Charge quantization	exp_09_harmonics.py	PART
Hydrogen Spectrum	Coulomb well orbital resonance	Bohr $\sim -1/n^2$	exp_10_hydrogen_spectrum.py	CONF
Spontaneous Emission	Zitterbewegung phase damping	n=2→1 decay	exp_10_hydrogen_spectrum.py	CONF

18 Conclusion

[STUB]

References

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